Towards Subseasonal-to-Seasonal Drought-to-Flood Prediction

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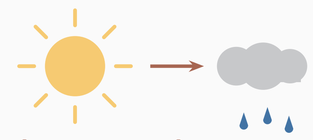
Introduction

Floods and droughts can occur in sequence — a **drought-to-flood transition (D2F)**. Globally, ∼**25% of floods are preceded by drought** [2], and socio-economic losses can be up to **8x greater** than from isolated hazards [3]. Forecasting D2Fs at subseasonal-to-seasonal lead times could support early action, but 2 research gaps remain ⟶

No catalogue of D2F events exists for Ireland

↓ Current focus

Subseasonal-to-seasonal **forecast skill** over Ireland has **not been assessed**



Subseasonal-to-seasonal D2F prediction

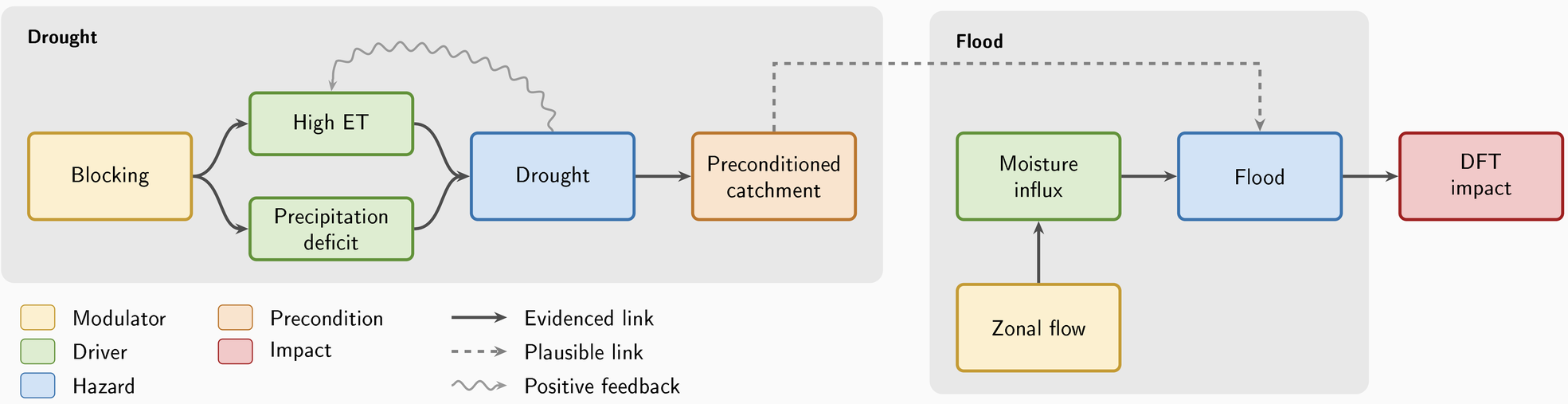


Fig. 1: Physical pathway for D2F events, after the framework of Bevacqua et al. [1].

Methods

As a first step towards D2F prediction, we verify 2 m temperature forecasts at **2–6 week** lead times, stratified by season (DJF, JJA) and lead. Mean bias is removed by **leave-one-year-out** correction, stratified by season and lead time. We also compare ERA5 against Met Éireann observations to understand our reference's own biases. **Rank histograms** show whether the ensemble is reliably spread; **Fair CRPSS** [4] measures probabilistic skill relative to climatology.

Data

ECMWF ENS sub-seasonal range (CY49R1)

11 members ▪ init. every 2 days ▪ weeks 2–6

2 m temperature ▪ 0.75° grid ▪ 2006–2016

Reference data

ERA5 ▪ Met Éireann 1 km gridded obs

Results

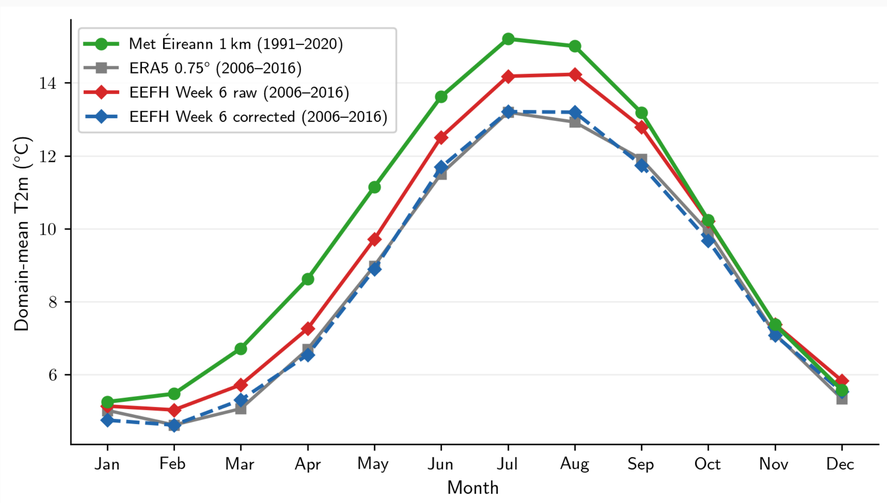


Fig. 2: Domain-mean seasonal cycle. ERA5 is cold-biased relative to Met Éireann, especially in JJA. Bias correction aligns the forecast with ERA5 but inherits this cold bias.

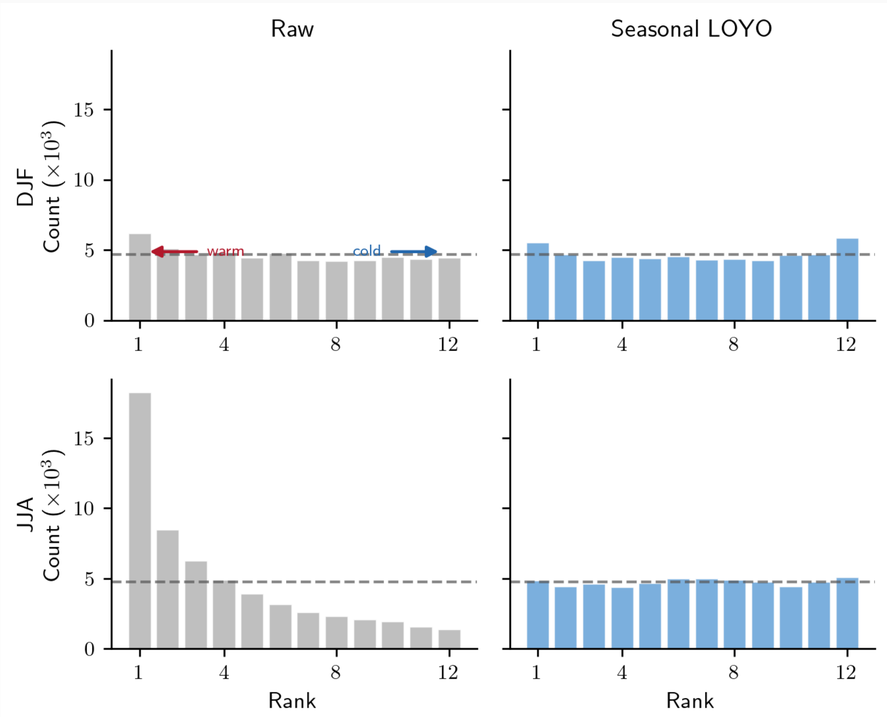


Fig. 3: Rank histograms (week 6). In JJA, raw forecasts are strongly warm-biased. In DJF, the ensemble is slightly overdispersive. Bias correction improves both, but introduces a slight cold bias from the ERA5 reference.

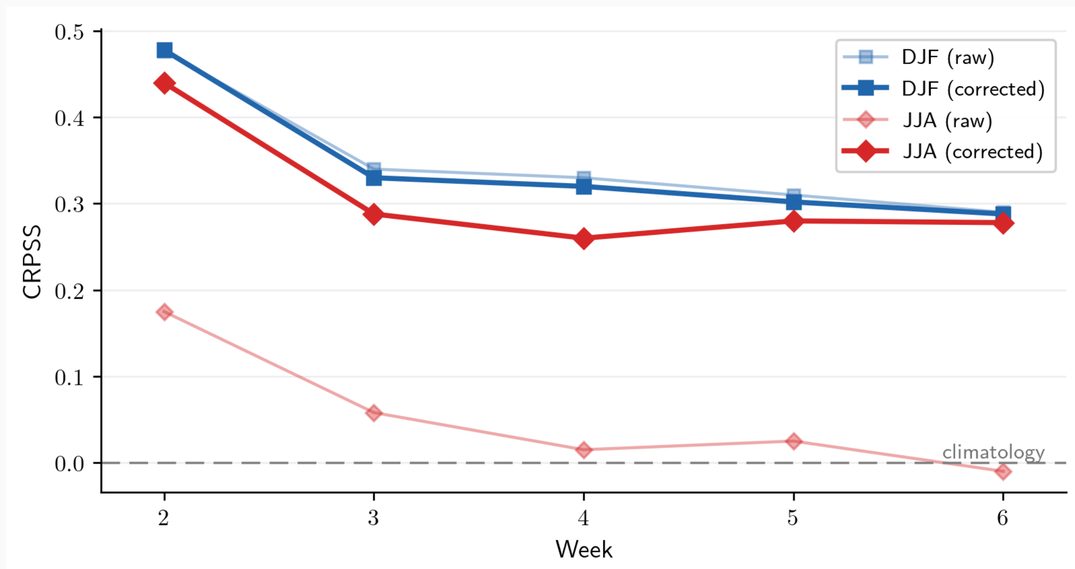


Fig. 4: Domain-mean Fair CRPSS by lead time and season. Raw JJA skill is near zero; bias correction recovers skill comparable to DJF. After correction, skill is positive to week 6 in both seasons.

- JJA warm bias is consistent with known IFS limitations in **soil moisture** and **vegetation–atmosphere coupling** [5]
- The forecast **outperforms the climatological mean** to week 6; **additional verification metrics** are needed for a complete assessment

Future Work

Compare with SEAS5 seasonal forecasts ⟶ Assess probabilistic verification metrics ⟶ Apply probabilistic bias correction

References

[1] Bevacqua et al. (2021). *Earth's Future*, 9, e2021EF002340.
[2] Matanó et al. (2024). *Environ. Res. Lett.*, 19, 064048.
[3] Worou & Messori (2025). *Environ. Res. Lett.*, 20, 104024.
[4] Ferro (2014). *Q. J. R. Meteorol. Soc.*, 140, 1917–1923.
[5] Haiden et al. (2018). *ECMWF Newsletter*, 157.